SANCHI GOYAL

Software Engineer | Full-Stack Developer | AI-Driven Applications | B.Tech Computer Science and Engineering
Portfolio: sanchi-goyal-github-io.netlify.app
Vellore, Tamil Nadu, India | +91-9452820926 | gsanchi21@gmail.com
linkedin.com/in/sanchi-goyal | github.com/sanzzzz-g | leetcode.com/u/sanzzzz_g

PROFESSIONAL SUMMARY

Software Engineer and B.Tech Computer Science student (CGPA 8.74/10) who builds full-stack and AI-driven products with React, TypeScript, Node.js, Express, and PostgreSQL. Independently built and shipped 4 deployed projects, including an ML demand-forecasting service and a hydroclimatic forecasting platform; merged an open-source pull request (GSSoC '26); solved 109+ LeetCode problems.

TECHNICAL SKILLS

Languages: Java, C++, C, Python, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks & Libraries: React.js, Vite, Node.js, Express.js, REST APIs, Tailwind CSS, shadcn/ui, Recharts, Leaflet.js, Chart.js
Databases: PostgreSQL, Supabase, SQL
Tools & Platforms: Git, GitHub, GitHub Actions (CI), Docker, Vercel, Netlify, Render, Google Cloud, VS Code
AI & ML: LLM agents (Google ADK, Gemini 2.5 Flash), tool calling (Google Search, MCP), scikit-learn, XGBoost, forecasting model evaluation
Core Concepts: Data Structures and Algorithms, OOP, System Design, DBMS, Database Design, Unit Testing, Agile/Scrum

PROJECTS

BloodBridge AI: Blood Supply Intelligence Platform *Solo Project | 2026*
Live Demo | GitHub

- Built a blood bank system tracking 8 blood groups across 5 components, with a FEFO allocation engine, ABO/Rh compatibility rules, atomic transactions preventing negative stock, a normalized PostgreSQL schema with migrations, and 4 access roles enforced via Supabase Row Level Security.
- Developed a FastAPI ML service forecasting blood demand at 7, 14, and 30-day horizons (Ridge Regression vs seasonal-naive baseline, 80/20 chronological split, MAE/RMSE) on synthetic demo data; wrote 12 automated test suites (Vitest, Pytest).

Tech: React 19, TypeScript, Supabase, PostgreSQL, Python, FastAPI, Docker, scikit-learn, Tailwind CSS

Marathwada Water Stress Index: Hydroclimatic Forecasting Platform *Solo Project | 2026*
Live Demo | GitHub | Data Pipeline

- Built a full-stack platform combining 6 indicators (rainfall, soil moisture, vegetation, temperature, groundwater, surface water) into a Water Stress Index for 8 districts across 2,112 district-months (2003-2024).
- Benchmarked 7 forecasting models on a chronological split (1,328 train / 272 validation / 368 test sequences); the Residual Ridge model reduced MAE by 30.1% versus a persistence baseline at 1 month, with 90% prediction intervals.
- Wrote 68 pytest tests and set up GitHub Actions CI (flake8 + pytest on every push and pull request) in the companion data-pipeline repo.

Tech: React 19, TypeScript, Vite, Tailwind CSS, Recharts, Leaflet, Express.js, Python, FastAPI, scikit-learn, XGBoost

Multi-Agent AI Systems with Google ADK *Guided Codelab | 2026*
GitHub

- Built 10 LLM agent implementations with Google ADK and Gemini 2.5 Flash (sequential, parallel, loop, routing, custom, agent-as-tool, memory), integrating Google Search and MCP tools.

Tech: Python, Google ADK, Gemini 2.5 Flash, MCP

Bharat Grid Intelligence: Power Grid Monitoring Dashboard *Solo Project | 2026*
Live Demo | GitHub

- Built a full-stack (React, Node.js, Express) grid monitoring dashboard with 8 REST API endpoints and a weighted 5-factor risk model (0-100 score) over a simulated 20-state energy dataset.
- Built a Leaflet.js map, 5 Chart.js charts, a 6-scenario simulator, an alert system (risk score above 80), and OpenWeather integration with 5-minute caching.

Tech: React, Vite, Node.js, Express.js, Leaflet.js, Chart.js, Axios

JalSuraksha: AI-Powered Water Security Platform *Solo Project | 2026*
Live Demo | GitHub

- Built a drought management platform (6 routed pages) with dataset upload in 7 file formats, Recharts dashboards, AI drought prediction, prediction history, multi-dataset joins, and Supabase auth; deployed on Render.

Tech: React, TypeScript, Vite, Tailwind CSS, shadcn/ui, Supabase, TanStack Query, Recharts

EXPERIENCE

Systems Engineer Intern | Northern Coalfields Limited (NCL), Systems Department *12 May 2026 - 26 Jun 2026*

- Audited and compiled an IT asset inventory of 1,906 PCs and UPS units across 19 projects and units; assessed 400 additional units for procurement toward a 2,306-unit planned fleet.
- Documented the live deployment of 7 SAP ERP modules (MM, FICO, SD, PP, PM, PS, HCM) and a 9-component CCMP cybersecurity stack (including SIEM, SOAR, EDR, UEBA, NTA, and IPS) in a detailed internship report.

EDUCATION

Vellore Institute of Technology, Vellore, Tamil Nadu, India *Aug 2024 - May 2028*
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.74/10.00

OPEN SOURCE & ACHIEVEMENTS

- **GSSoC '26 Open-Source Contributor:** merged a bug-fix PR in DevTrack (PR #1200) fixing 404 errors on mixed-case usernames via case-insensitive queries and canonical redirects; built an incremental analysis feature that caches metadata to avoid redundant re-scans.
- **LeetCode:** solved 109+ problems in Java, C, and C++ (Arrays, Dynamic Programming, Hash Tables). Profile

CERTIFICATIONS

Google Cloud: Engineer AI Agents with ADK; Basics of Compute Skill Badge | Cisco: Networking Basics